

ECSE-4040

Digital Electronics

Lecture 7

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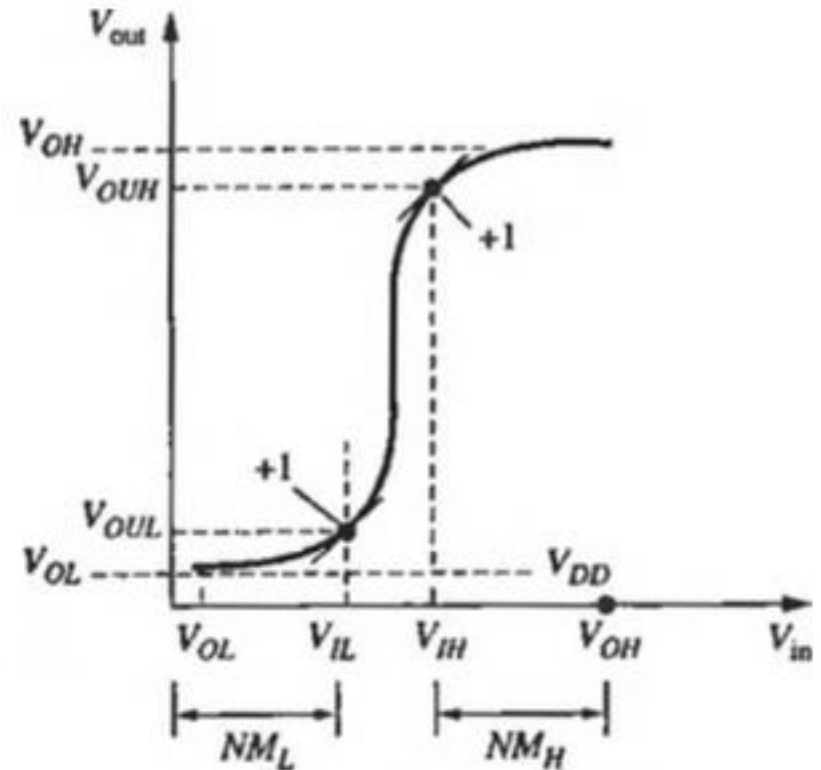
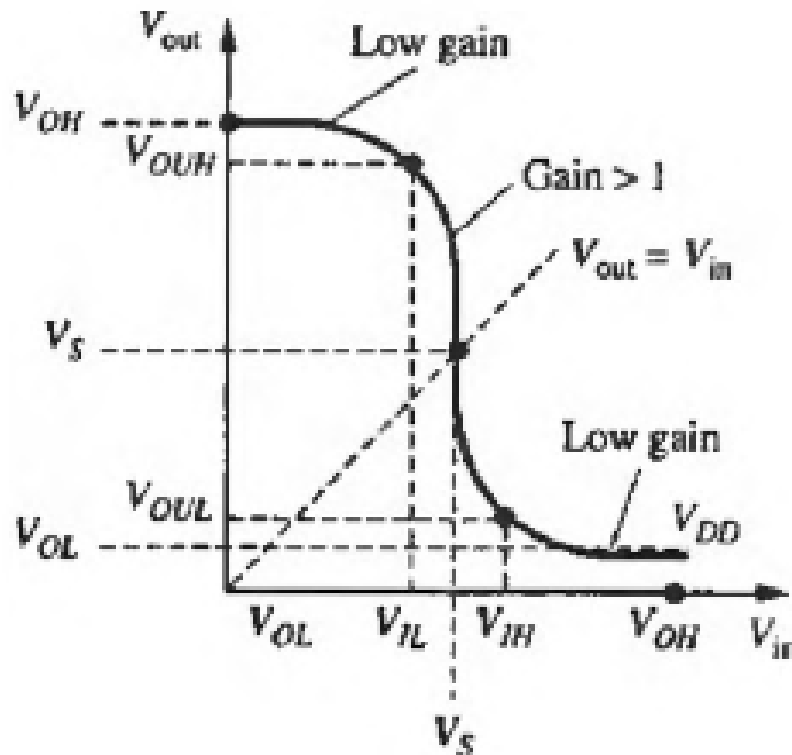


Lecture Outline

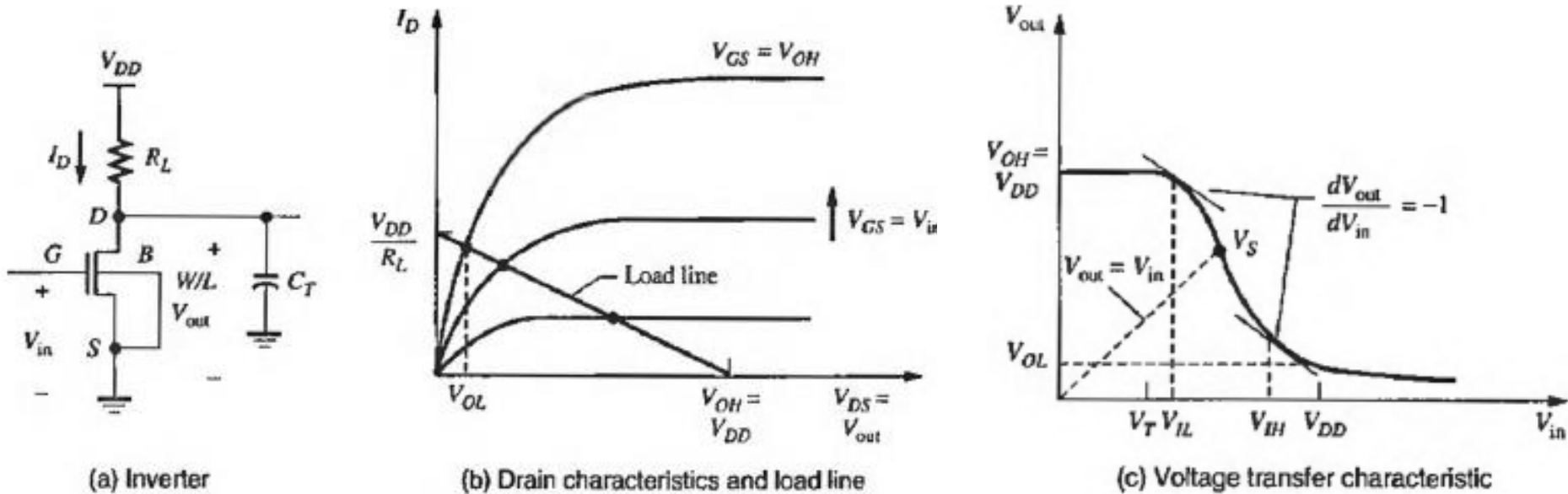
- Inverter vs. Buffer
- NMOS Inverters
- CMOS Inverter
 - VTC Critical Points
 - Inverter Sizing
- Pseudo-NMOS Inverter



Inverter vs. Buffer



Resistive-Load Inverter Design

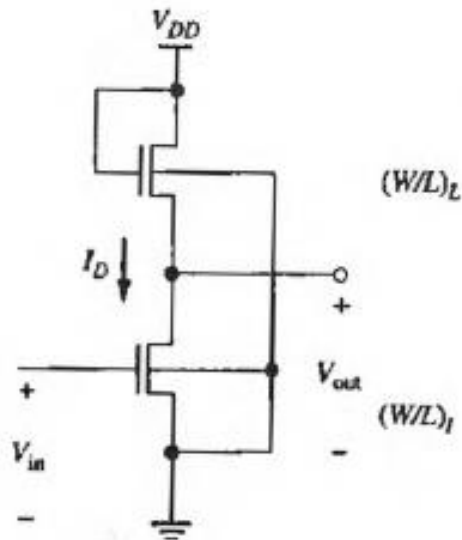


VTC Critical Points:

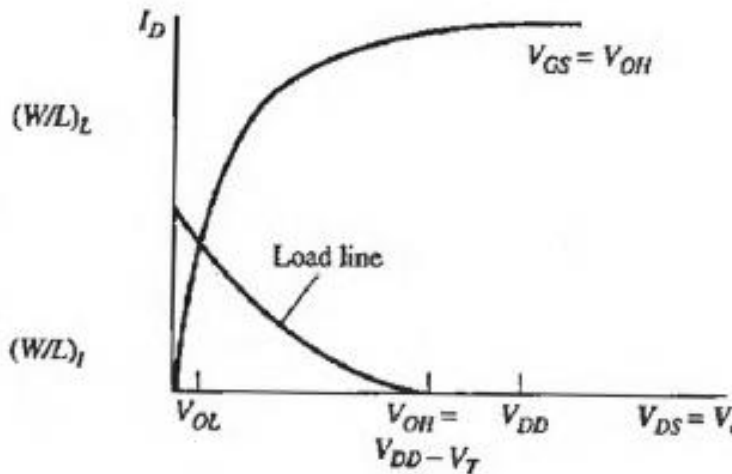
V_{OH} , V_{OL} , V_{IL} , V_{OL} , V_S

Midpoint of VTC: $V_{in} = V_{out} = V_S$

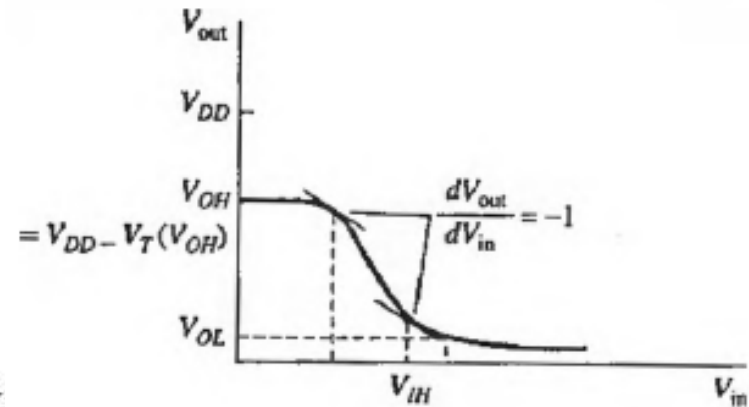
Saturated Enhancement-Load Inverter



(a) Enhancement-loaded inverter, saturated load



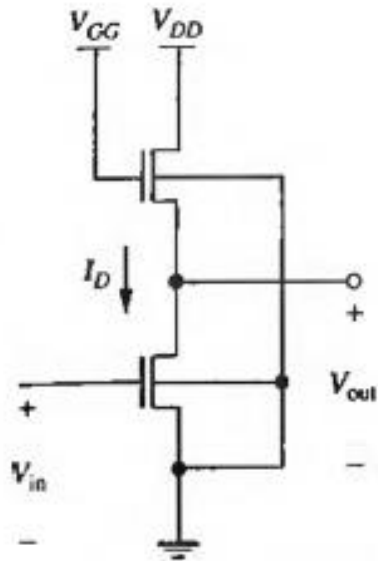
(b) Drain characteristic and load line



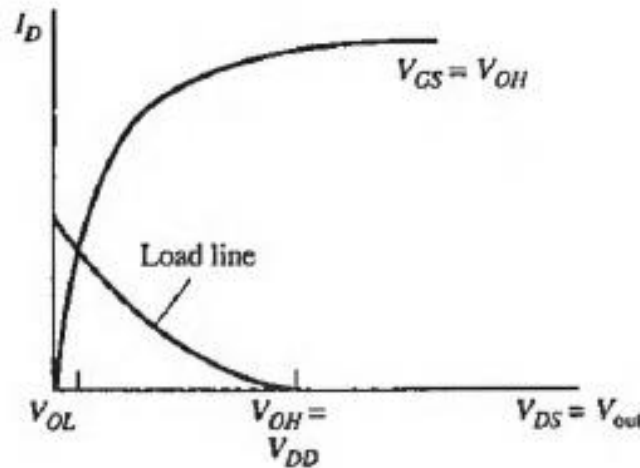
(c) Voltage transfer characteristic

- Single NMOSFET with gate and drain connected as load or pullup device
- **Drawback:** $V_{OH} \neq V_{DD}$

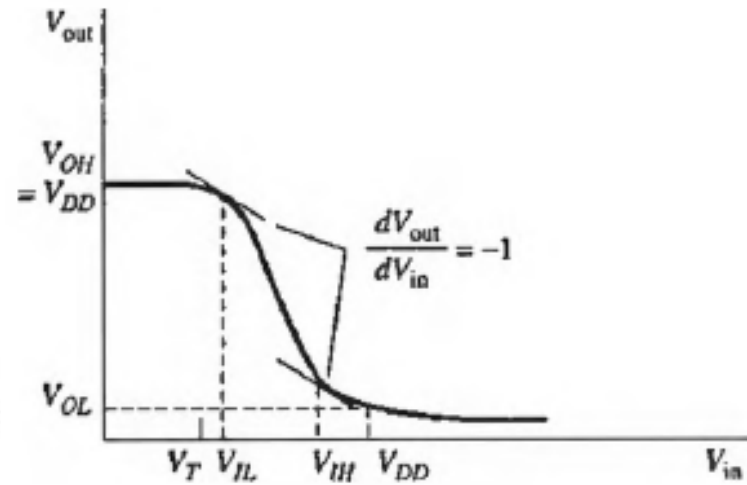
Linear Enhancement-Load Inverter



(a) Enhancement-loaded inverter, linear load



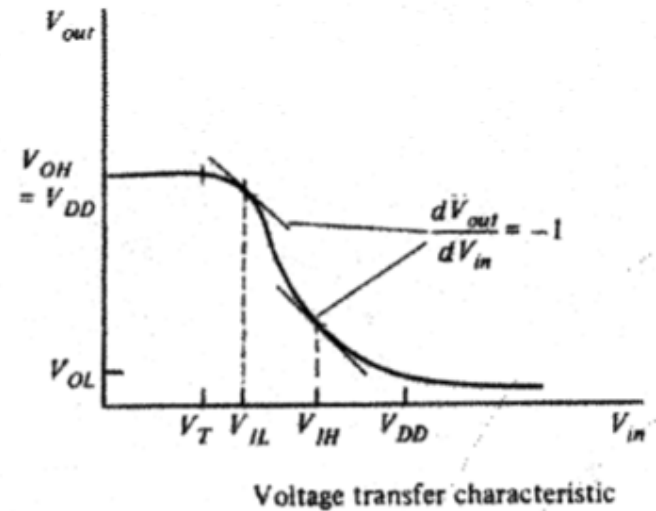
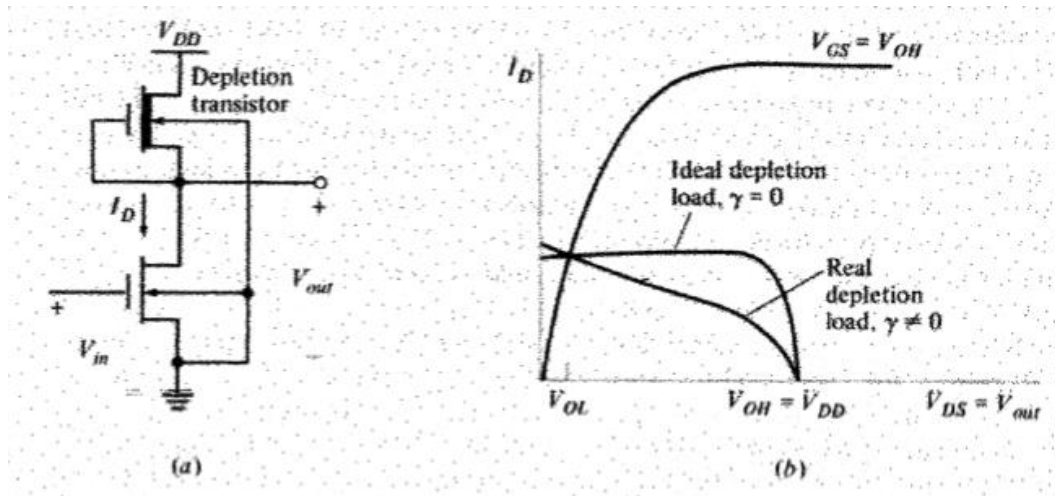
(b) Drain characteristic and load line



(c) Voltage transfer characteristic

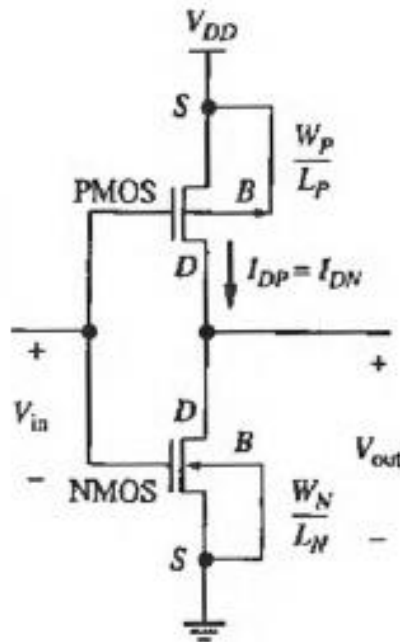
- Single NMOSFET with gate voltage above V_{DD} as load or pullup device

Depletion Load Inverter

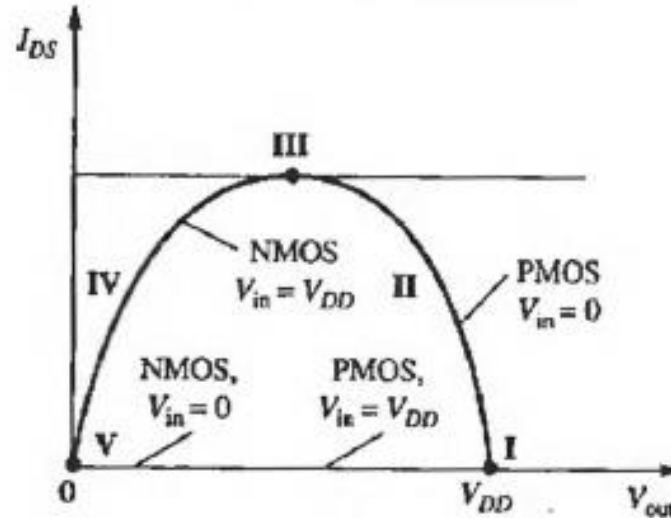


- Single depletion-mode NMOSFET as load or pullup device

CMOS Inverter

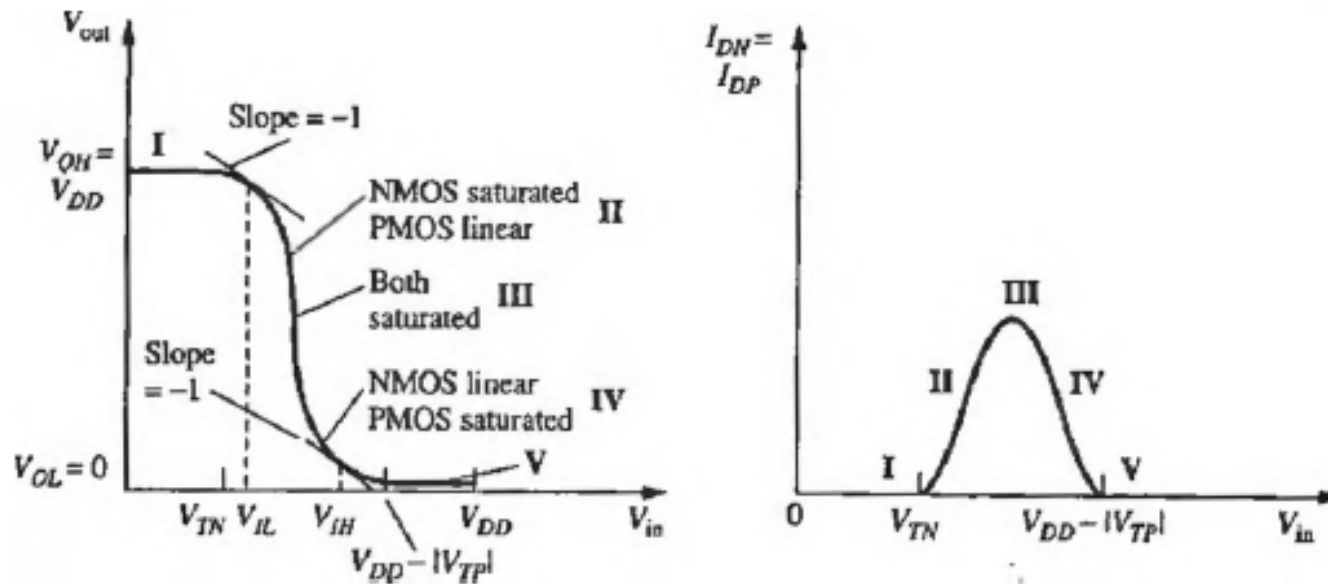


(a) Inverter



(b) Load line

CMOS Inverter



(c) Voltage transfer characteristic and current

CMOS Inverter

Operation of CMOS Inverter:

1. Region I: $0 V < V_{in} < V_{TN}$ (n-off, p-lin)

NMOS Cutoff ($V_{GS} < V_{TN}$)

PMOS conducting, but little current ($V_{GS} = -V_{DD}$)

So, $V_{out} \sim V_{OH} = V_{DD}$

2. Region II: $V_{in} = V_{IL} > V_{TN}$ (n-sat, p-lin)

NMOS turns on and is in saturation

($V_{DS} \sim V_{DD}$, $V_{GS} - V_{TN} > 0$ but $\ll V_{DD}$)

PMOS is in linear region

($|V_{GS} - V_{TP}| < V_{DD}$, $V_{DS} \sim 0$)

CMOS Inverter

$$I_{DN}(\text{sat}) = I_{DP}(\text{lin})$$

$$\frac{W_N \mu_{\text{sat}} C_{\text{ox}} (V_{\text{in}} - V_{TN})^2}{(V_{\text{in}} - V_{TN}) + E_{CN} L_N}$$

$$= \frac{W_P}{L_P} \frac{\mu_P C_{\text{ox}}}{\left(1 + \frac{V_{DD} - V_{\text{out}}}{E_{CP} L_P}\right)} \left[(V_{DD} - V_{\text{in}} - |V_{TP}|)(V_{DD} - V_{\text{out}}) - \frac{(V_{DD} - V_{\text{out}})^2}{2} \right]$$

3. Region III: $V_{\text{in}} = V_S$ (n-sat, p-sat)

Both FETs in saturation

$$(|V_{DS}| \sim V_{DD}/2 \sim V_{DD}/2 - V_{TN} \text{ (or } |V_{TP}|))$$

$$\sim V_S - V_{TN} \text{ or } (|V_{TP}|)$$

CMOS Inverter

$$I_{DN}(\text{sat}) = I_{DP}(\text{sat})$$

$$\frac{W_N \mu_{\text{sat}} C_{\text{ox}} (V_{\text{in}} - V_{TN})^2}{(V_{\text{in}} - V_{TN}) + E_{CN} L_N} = \frac{W_P \mu_{\text{sat}} C_{\text{ox}} (V_{DD} - V_{\text{in}} - |V_{TP}|)^2}{(V_{DD} - V_{\text{in}} - |V_{TP}|) + E_{CP} L_P}$$

Setting $V_{\text{in}} = V_S \sim V_{DD}/2$ gives

$$\frac{W_N \mu_{\text{sat}} C_{\text{ox}} (V_S - V_{TN})^2}{(V_S - V_{TN}) + E_{CN} L_N} = \frac{W_P \mu_{\text{sat}} C_{\text{ox}} (V_{DD} - V_S - |V_{TP}|)^2}{(V_{DD} - V_S - |V_{TP}|) + E_{CP} L_P}$$

$$\frac{W_N (V_S - V_{TN})^2}{E_{CN} L_N} = \frac{W_P (V_{DD} - V_S - |V_{TP}|)^2}{E_{CP} L_P}$$

Then,

$$V_S = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi} \quad \text{where} \quad \chi = \sqrt{\frac{\frac{W_N}{E_{CN} L_N}}{\frac{W_P}{E_{CP} L_P}}} = \sqrt{\frac{\mu_n W_N}{\mu_p W_P}}$$

HJS Ex. 4.6

Computation of V_S for CMOS Inverter in $0.13\ \mu\text{m}$ Technology

Problem:

Calculate the switching voltage of a CMOS inverter with each the following PMOS sizes: $W_p = 400\ \text{nm}$ and $W_p = 100\ \text{nm}$. W_n is the same size of $100\ \text{nm}$ for both cases. How does change in W_p affect the switching voltage?

Solution:

Compute V_S using Equations (4.14) and (4.15):

$$W_n = 0.1\ \mu\text{m} \quad W_p = 0.4\ \mu\text{m}$$

$$X = \sqrt{\frac{\frac{W_n}{E_{CN}L_n}}{\frac{W_p}{E_{CP}L_p}}} = \sqrt{\frac{W_n E_{CP}}{W_p E_{CN}}} = \sqrt{\frac{(0.1)(24)}{(0.4)(6)}} = 1.0$$

$$V_S = \frac{0.8 + (0.4)1.0}{1 + 1.0} = 0.6\ \text{V}$$

HJS Ex. 4.6

Re-computing for the second case:

$$W_N = 0.1 \mu\text{m} \quad W_P = 0.1 \mu\text{m}$$

$$X = \sqrt{\frac{\frac{W_N}{E_{CN}L_N}}{\frac{W_P}{E_{CP}L_P}}} = \sqrt{\frac{W_N E_{CP}}{W_P E_{CN}}} = \sqrt{\frac{(0.1)(24)}{(0.1)(6)}} = 2.0$$

$$V_S = \frac{0.8 + (0.4)2.0}{1 + 2.0} = 0.53 \text{ V}$$

We find that lowering W_P decreases V_S which shifts the VTC to the left.

CMOS Inverter

4. Region IV: $V_{in} = V_{IH} < V_{DD} - |V_{TP}|$ (n-lin, p-sat)

PMOS turns on and is in saturation

($|V_{DS}| \sim V_{DD}$, $|V_{GS} - |V_{TP}|| > 0$ but $\ll V_{DD}$)

NMOS is in linear region

($V_{GS} - V_{TN} < V_{DD}$, $V_{DS} \sim 0$)

$I_{DN}(\text{lin}) = I_{DP}(\text{sat})$

$$\frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_N}\right)} \left[(V_{in} - V_{TN}) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - V_{in} - |V_{TP}|)^2}{(V_{DD} - V_{in} - |V_{TP}|) + E_{CP} L_P}$$

CMOS Inverter

Operation of CMOS Inverter:

5. Region V: $V_{DD} > V_{in} > V_{DD} - |V_{TP}|$ (n-lin, p-off)

PMOS Cutoff ($V_{GS} < |V_{TP}|$)

NMOS conducting, but little current ($V_{GS} = V_{DD}$)

So, $V_{out} \sim 0$ V

CMOS Inverter

Noise Margins:

$$V_{OH} = V_{DD} \text{ and } V_{OL} = 0 \text{ V}$$

At $V_{in} = V_{IL}$ (Region II), $I_{DN}(\text{sat}) = I_{DP}(\text{lin})$

$$\frac{W_N \mu_{sat} C_{ox} (V_{in} - V_{TN})^2}{(V_{in} - V_{TN}) + E_{CN} L_N} = \frac{W_P}{L_P} \frac{\mu_p C_{ox}}{\left(1 + \frac{V_{DD} - V_{out}}{E_{CP} L_P}\right)} \times \left[(V_{DD} - V_{in} - |V_{TP}|)(V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right]$$

$$\frac{W_N \mu_{sat} C_{ox} (V_{in} - V_{TN})^2}{E_{CN} L_N} = \frac{W_P}{L_P} \mu_p C_{ox} \times \left[(V_{DD} - V_{in} - |V_{TP}|)(V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right]$$

CMOS Inverter

Differentiating with respect to V_{in} gives

$$\frac{W_N}{L_N} \mu_n (V_{in} - V_{TN}) = \frac{W_P}{L_P} \mu_p \left[(V_{DD} - V_{in} - |V_{TP}|) \left(-\frac{\partial V_{out}}{\partial V_{in}} \right) + (V_{DD} - V_{out})(-1) - (V_{DD} - V_{out}) \left(-\frac{\partial V_{out}}{\partial V_{in}} \right) \right]$$

Setting $k_N, k_P, \frac{\partial V_{out}}{\partial V_{in}} = -1.0$ and $V_{in} = V_{IL}$ gives

$$V_{IL} = \frac{2V_{out} - V_{DD} - |V_{TP}| + (k_N/k_P)(V_{TN})}{1 + (k_N/k_P)}$$

CMOS Inverter

At $V_{in} = V_{IH}$ (Region IV), $I_{DN}(lin) = I_{DP}(sat)$

$$\frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_N}\right)} \left[(V_{in} - V_{TN}) V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - V_{in} - |V_{TP}|)^2}{(V_{DD} - V_{in} - |V_{TP}|) + E_{CP} L_P}$$

$$\frac{W_N}{L_N} \mu_n C_{ox} \left[(V_{in} - V_{TN}) (V_{out}) - \frac{(V_{out})^2}{2} \right] = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - V_{in} - |V_{TP}|)^2}{E_{CP} L_P}$$

$$\frac{W_N}{L_N} \mu_n C_{ox} \left[(V_{in} - V_{TN}) (V_{out}) - \frac{(V_{out})^2}{2} \right] = \frac{W_P}{L_P} \frac{\mu_p C_{ox}}{2} (V_{DD} - V_{in} - |V_{TP}|)^2$$

CMOS Inverter

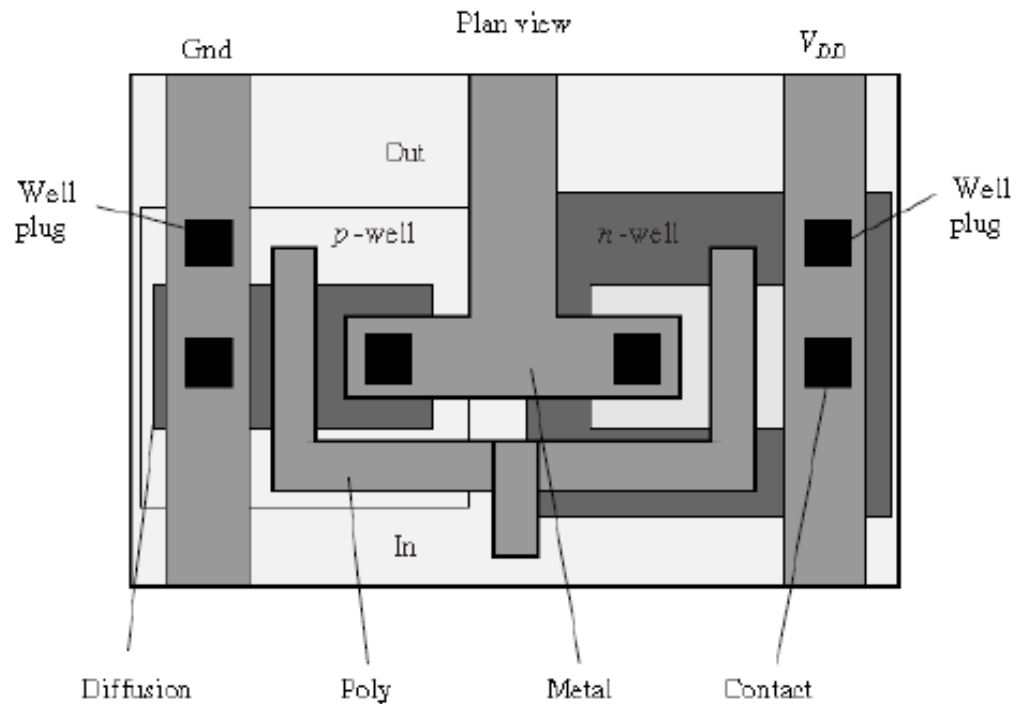
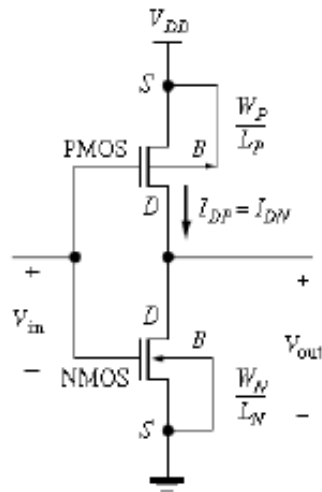
Differentiating with respect to V_{in} gives

$$\begin{aligned} \frac{W_N}{L_N} \mu_n C_{ox} \left[(V_{in} - V_{TN}) \frac{\partial V_{out}}{\partial V_{in}} + (V_{out} - V_{out}) \frac{\partial V_{out}}{\partial V_{in}} \right] \\ = \frac{W_P}{L_P} \mu_p C_{ox} (V_{DD} - V_{in} - |V_{TP}|)(-1) \end{aligned}$$

Setting $k_N, k_P, \frac{\partial V_{out}}{\partial V_{in}} = -1.0$ and $V_{in} = V_{IH}$ gives

$$V_{IH} = \frac{2V_{out} + V_{TN} + (k_P/k_N)(V_{DD} - |V_{TP}|)}{1 + (k_P/k_N)}$$

CMOS Inverter Layout



Note Minimum size, well-plugs, design rules

HJS Ex. 4.7

SPICE Analysis and Hand Calculation of CMOS Inverter in 0.18 μm Technology

Problem:

Compare SPICE analysis and hand analysis for a CMOS inverter of Figure 4.12a, with $W_N = 400 \text{ nm}$ and $W_P = 800 \text{ nm}$, for the VTC parameters. Use 0.18 μm technology parameters as follows:

$$\mu_n = 270 \text{ cm}^2/\text{V}\cdot\text{s}, \quad \mu_p = 70 \text{ cm}^2/\text{V}\cdot\text{s},$$

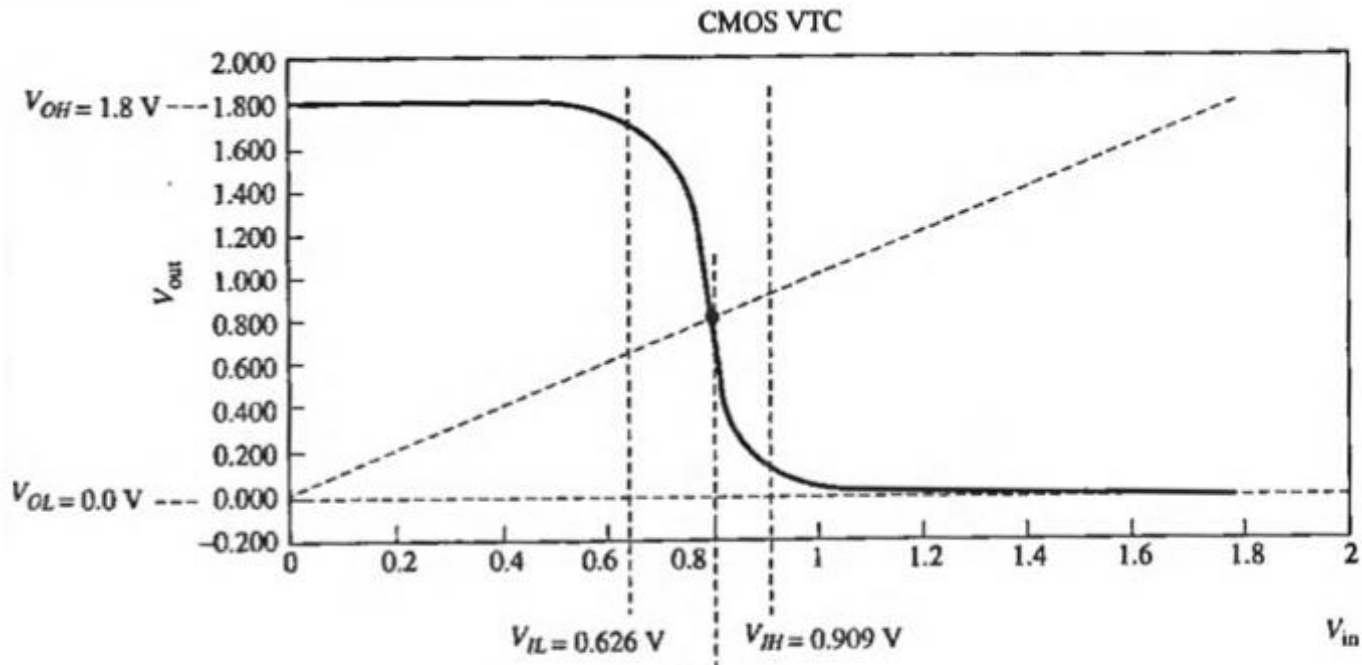
$$C_{ox} = 1.0 \text{ fF}/\text{cm}^2, \quad V_{TN} = 0.5 \text{ V}, \quad V_{TP} = -0.5 \text{ V}$$

$$E_{CN}L = 1.2 \text{ V}, \quad E_{CP}L = 4.8 \text{ V}, \quad v_{sat} = 8 \times 10^6 \text{ cm/s},$$

$$V_{DD} = 1.8 \text{ V}, \quad L = 200 \text{ nm}$$

HJS Ex. 4.7

Solution:



The VTC parameters from SPICE are $V_{OH} = 1.8\text{ V}$, $V_{OL} = 0\text{ V}$, $V_{IL} = 0.626\text{ V}$, $V_{IH} = 0.909\text{ V}$, $V_S = 0.8\text{ V}$. Therefore, the noise margins are

$$NM_H = 1.8 - 0.909 = 0.891\text{ V}$$

$$NM_L = 0.626 - 0 = 0.626\text{ V}$$

HJS Ex. 4.7

For hand analysis, we already know that $V_{OH} = 1.8 \text{ V}$, and $V_{OL} = 0 \text{ V}$.

The computation of V_S can be carried out as before, with $0.18 \mu\text{m}$ parameters:

$$W_N = 0.4 \mu\text{m} \quad W_P = 0.8 \mu\text{m}$$

$$X = \sqrt{\frac{\frac{W_N}{E_{CN}L_N}}{\frac{W_P}{E_{CP}L_P}}} = \sqrt{\frac{W_N E_{CP}}{W_P E_{CN}}} = \sqrt{\frac{(0.4)(24)}{(0.8)(6)}} = 1.41$$

$$V_S = \frac{1.8 - 0.5 + (0.5)1.41}{1 + 1.41} = 0.83 \text{ V}$$

This value is quite accurate relative to the SPICE results. The values for V_{IL} and V_{IH} require some effort but eventually produce $V_{IL} \approx 0.7 \text{ V}$ and $V_{IH} \approx 1.0 \text{ V}$. Therefore,

$$NM_H = 1.8 - 1.0 = 0.8 \text{ V}$$

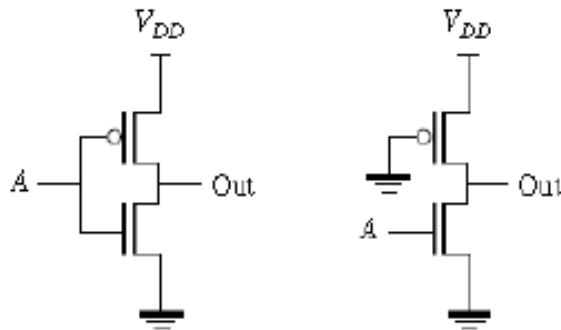
$$NM_L = 0.7 - 0 = 0.7 \text{ V}$$

While these are not accurate results, they are within 15% of SPICE and are good enough for hand analysis.

Pseudo-NMOS Inverter

To address issues with NMOS loads

- Saturated NMOS load, a.k.a diode connected load, has degraded V_{OH}
- Linear NMOS load requires two supplies and extra area/interconnects
- CMOS gates require multiple loads for multi fanin inputs (?)



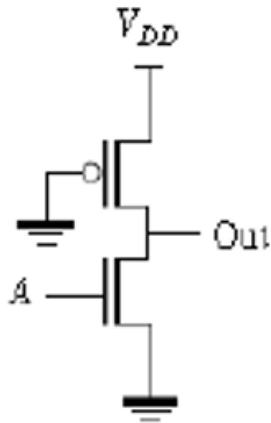
Solves the abovementioned problems, However

- *Ratioed* inverter, i.e. V_{OL} a function of two device ratios
- Large T_{PLH} (why?)

$$I_{DP}(\text{sat}) = I_{DN}(\text{lin})$$

$$\frac{W_P \mu_{\text{sat}} C_{ox} (V_{DD} - |V_{TH}|)^2}{(V_{DD} - |V_{TH}|) + E_{CP} L_P} = \frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{OL}}{E_{CN} L_N}\right)} \left[(V_{DD} - V_{TN})(V_{OL}) - \frac{(V_{OL})^2}{2} \right] \Rightarrow V_{OL} = \frac{I_{DP}(\text{sat})}{k_N (V_{DD} - V_{TN})}$$

Pseudo-NMOS Inverter



V_{in} goes from 0 to V_{DD}

1. Region I: $V_{in} > V_{TN}$; $V_{DD} \gg V_{in}$

$$I_{DP}(\text{lin}) = I_{DN}(\text{sat})$$

2. Region II: $V_{in} = V_{DD}$; $V_{out} = V_{OL} \sim 0 < |V_{TP}|$

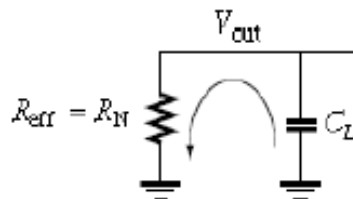
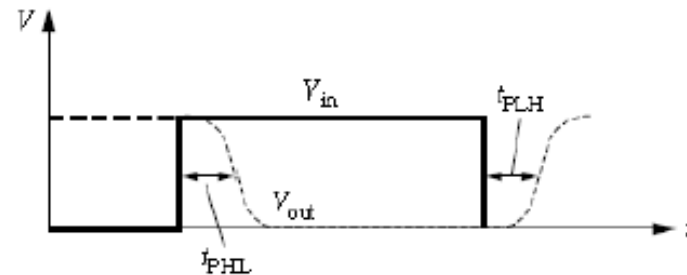
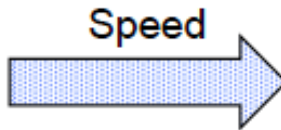
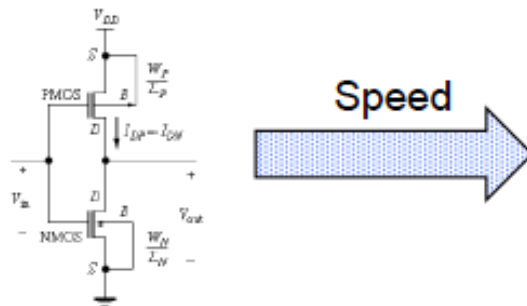
$$I_{DP}(\text{sat}) = I_{DN}(\text{lin})$$

$$\frac{W_P \mu_{sat} C_{ox} (V_{DD} - |V_{TP}|)^2}{(V_{DD} - |V_{TP}|) + E_{CP} L_P} = \frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{OL}}{E_{CN} L_N}\right)} \left[(V_{DD} - V_{TN})(V_{OL}) - \frac{(V_{OL})^2}{2} \right]$$

$$\Rightarrow V_{OL} = \frac{I_{DP}(\text{sat})}{k_N (V_{DD} - V_{TN})}$$

Inverter Sizing

- In ratioed logic families (such as diode-connected load or pseudo NMOS, V_{OL} is a priority so the Size of load is mainly determined by the choice of V_{OL}
- In non-ratioed logic families (such CMOS inverters) propagation delay is important



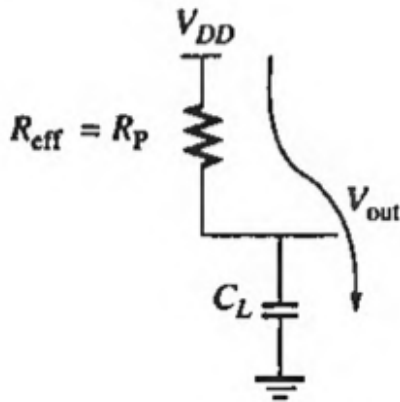
Pull-down (output capacitor discharge)
through the NMOS transistor

$$V_{out}(t) = V_{DD} e^{-t/R_N C_L}$$

T_{PHL} is the time it take for the capacitor to discharge to 50% of the final value

$$\tau = 0.69 R_{eff} C_L$$

Inverter Sizing



$$V_{\text{out}}(t) = V_{DD}(1 - e^{-t/R_P C_L})$$

Pull-up (output capacitor charging)
through the PMOS transistor

Inverter Sizing

- The value of T_{PHL} or T_{PLH} determine the value of the resistor (average on-resistance)
- Note that in the pull-down case (NMOS on) device starts in saturation (V_{DD} across its V_{DS} and might enter triode at some point, therefore it makes sense to define an effective resistance

$$R_{eqn} = 12.5 \text{ k}\Omega/\square$$

$$R_{eqp} = 30 \text{ k}\Omega/\square$$

Unit resistance of the devices (per square)

That is for a device with $L=W$ (obtained from SPICE simulation), note the ration 2.4, does it remind of a number?

$$R_N = R_{eqn} \times \frac{L_N}{W_N}$$

$$R_P = R_{eqp} \times \frac{L_P}{W_P}$$

Total Resistance inversely proportional to W/L (remember that in triode region we could model the transistor simply with a resistor!

Review Ex. 4.9 of HJS

HJS Ex. 4.9

Comparison of CMOS and Pseudo-NMOS Inverters

Problem:

Given the CMOS and pseudo-NMOS inverters of Figure 4.14, size the two inverters to achieve the following specifications, assuming that the output loading is 50 fF. Use 0.13 μm technology parameters:

$$\mu_n = 270 \text{ cm}^2/\text{V-s}, \quad \mu_p = 70 \text{ cm}^2/\text{V-s}, \quad C_{ox} = 1.6 \text{ } \mu\text{F}/\text{cm}^2,$$

$$V_{TN} = 0.4 \text{ V}, \quad V_{TP} = -0.4 \text{ V}$$

$$E_{CNL} = 0.6 \text{ V}, \quad E_{CPL} = 2.4 \text{ V}, \quad v_{sat} = 8 \times 10^6 \text{ cm/s},$$

$$V_{DD} = 1.2 \text{ V}, \quad L = 100 \text{ nm}$$

CMOS Inverter: $t_{PHL} = t_{PLH} < 50 \text{ ps}$; $V_{OH} = 1.2 \text{ V}$ and $V_{OL} = 0 \text{ V}$; low dc power; minimum area

Pseudo-NMOS Inverter: $t_{PHL} < 50 \text{ ps}$; $V_{OH} = 1.2 \text{ V}$ and $V_{OL} = 0.1 \text{ V}$; minimum dc power; minimum area

HJS Ex. 4.9

Solution:

For the CMOS inverter, timing is the highest priority. The noise margin parameters of V_{OH} and V_{OL} are satisfied by choosing $V_{DD} = 1.2$ V. The power requirement is satisfied since it is a standard CMOS gate. The area requirement can be satisfied by meeting but not exceeding the timing specification. That is, we do not size the devices any larger than needed. For the timing requirement, we make use of (4.22c):

$$t = 0.7R_{\text{eff}}C_L = 0.7R_{\text{eff}} \times 50 \text{ fF} = 50 \text{ ps} \Rightarrow R_{\text{eff}} = 1.4 \text{ k}\Omega$$

Using (4.24a), we can determine the NMOS (W/L) value:

$$R_N = R_{\text{eqn}} \times \frac{L}{W} = 12.5 \text{ k}\Omega \times \frac{L}{W} = 1.4 \text{ k}\Omega \Rightarrow \frac{W_N}{L_N} = 8.75$$

Now, (4.24b) can be used to obtain the value of the PMOS device sizes:

$$R_P = R_{\text{eqp}} \times \frac{L}{W} = 30 \text{ k}\Omega \times \frac{L}{W} = 1.4 \text{ k}\Omega \Rightarrow \frac{W_P}{L_P} = 21$$

Since $L = 0.1 \text{ }\mu\text{m}$, we now have the sizes that satisfy the design specifications:

$$\frac{W_N}{L_N} = \frac{0.875 \text{ }\mu\text{m}}{0.1 \text{ }\mu\text{m}} \quad \frac{W_P}{L_P} = \frac{2.1 \text{ }\mu\text{m}}{0.1 \text{ }\mu\text{m}}$$

HJS Ex. 4.9

For the pseudo-NMOS gate, we prioritize V_{OL} in the sizing process since it is a ratioed circuit. We can satisfy the V_{OH} requirement by choosing $V_{DD} = 1.2\text{ V}$. The sizing of the NMOS device should be the same as in the CMOS case since its timing requirement is the same. We can satisfy the power and area requirements by meeting but not exceeding the V_{OL} and t_{PHL} requirements. To compute the PMOS device size, use (4.21):

$$\frac{W_P V_{sat} C_{ox} (V_{DD} - |V_{TP}|)^2}{(V_{DD} - |V_{TP}|) + E_{CP} L_P} = \frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{OL}}{E_{CN} L_N}\right)} \left[(V_{DD} - V_{TN})(V_{OL}) - \frac{(V_{OL})^2}{2} \right]$$

Using this equation, we apply the known values and solve for W_P :

$$\begin{aligned} & \frac{W_P 8(10^6)(1.6)(10^{-6})(1.2 - 0.4)^2}{(1.2 - 0.4) + 2.4} \\ &= 8.75 \times \frac{430 \mu\text{A/V}^2}{\left(1 + (0.065/0.6)\right)} \left[(1.2 - 0.4)(0.065) - \frac{(0.065)^2}{2} \right] \\ &\therefore W_P = 0.665 \mu\text{m} \end{aligned}$$

HJS Ex. 4.9

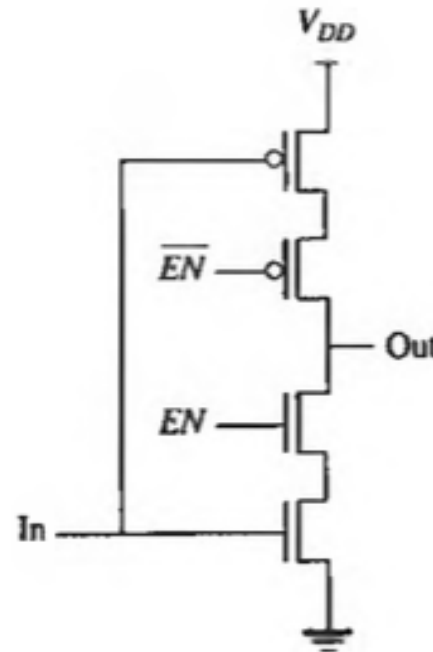
Based on this result, we can specify the device sizes:

$$\frac{W_N}{L_N} = \frac{0.875 \mu\text{m}}{0.1 \mu\text{m}} \quad \frac{W_P}{L_P} = \frac{0.665 \mu\text{m}}{0.1 \mu\text{m}}$$

For the pseudo-NMOS case, the two device sizes are about the same. This implies that t_{PLH} will be well above 100 ps (i.e., more than twice as large as t_{PHL}). This is the price that must be paid for using this type of load. There will also be more power dissipated with the output low as in the NMOS load cases. However, the area is far less than CMOS for high-fanin gate circuits, as will be seen later.

Tristate Buffer

- Besides high or low, a 3rd high impedance state is possible at output
- 2 additional transistors that can be enabled with EN and its complement
- When EN is high, the 2 transistors are on, the buffer functions as a regular inverter
- When EN is low, both transistors are off and output is at high impedance condition (called *high Z state*)
- Inverter no longer control output voltage



IN	EN	OUT
0	0	Z
1	0	Z
0	1	1
1	1	0

Truth Table

Summary

- Resistive load inverter has a challenging trade-off among different design criteria such noise margin, area and power
- Diode-connected inverter is *ratioed* and suffers from degraded V
- Linear NMOS load is *ratioed* and suffers from large area and several power connections
- CMOS Inverter is not ratioed, has low power consumption and can be optimized for certain propagation delay, however suffers from several loads when used for multi input gates
- Pseudo-NMOS Inverter is *ratioed*, uses single PMOS load and may have large propagation delay
- Going forward our focus will be on CMOS Gates

